The Cross Product

Name: _____ Date: _____

Definition. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$. Their *cross product* is

$$\mathbf{u} \times \mathbf{v} = \langle _____, _____, _____ \rangle .$$

Unlike the dot product, the cross product produces a _____. It is defined here only for vectors in $\mathbb{R}^3$.

Note. The determinant notation is a compact mnemonic:

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = _____ - _____ + _____ .$$

The minus sign in the middle cofactor is a common source of errors.

Example. Let $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle 0, 4, 1 \rangle$. Compute $\mathbf{u} \times \mathbf{v}$, and verify that the result is orthogonal to both vectors.

Work:

Theorem. If θ is the angle between nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, where $0 \le \theta \le \pi$, then

$$\|\mathbf{u} \times \mathbf{v}\| = _____.$$

The vector $\mathbf{u} \times \mathbf{v}$ is orthogonal to both _____ and _____.

Its direction is determined by the _____.

Commentary. Point the fingers of your right hand along $\mathbf{u}$ and curl them toward $\mathbf{v}$. Your thumb points in the direction of $\mathbf{u} \times \mathbf{v}$. Reversing the order reverses the direction.

Corollary. For nonzero vectors,

$$\mathbf{u} \times \mathbf{v} = \mathbf{0} \iff _____.$$

Theorem. For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ and scalar c:

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= _____ && \text{(anticommutative)}, \\ \mathbf{u} \times (\mathbf{v} + \mathbf{w}) &= _____, \\ (c\mathbf{u}) \times \mathbf{v} &= _____, \\ \mathbf{u} \times \mathbf{u} &= _____. \end{aligned}$$

Note. The standard unit vectors follow the cyclic pattern

$$\mathbf{i} \times \mathbf{j} = _____, \qquad \mathbf{j} \times \mathbf{k} = _____, \qquad \mathbf{k} \times \mathbf{i} = _____.$$

Reversing any product changes its sign.

Example. Without using the component formula, simplify

$$(2\mathbf{i} - \mathbf{k}) \times (3\mathbf{j} + 4\mathbf{k}).$$

Work:

Example. Find two unit vectors orthogonal to both $\mathbf{a} = \langle 1, 2, -1 \rangle$ and $\mathbf{b} = \langle 3, 0, 2 \rangle$.

Work:

Theorem. If $\mathbf{u}$ and $\mathbf{v}$ form adjacent sides of a parallelogram, then

$$\text{Area}_{\text{parallelogram}} = _____, \qquad \text{Area}_{\text{triangle}} = _____.$$

Example. Find the area of the triangle with vertices $P = (1, 0, 2)$, $Q = (3, 1, -1)$, and $R = (0, 4, 1)$.

First choose two sides that begin at the same vertex:

$$\overrightarrow{PQ} = _____, \qquad \overrightarrow{PR} = _____.$$

Work:

Note. For a triangle, the vectors must represent two sides meeting at one vertex. The order of the cross product changes direction, but not area.

Definition. Let $\mathbf{r}$ be the position vector from a pivot point to the point where a force $\mathbf{F}$ is applied. The *torque* about the pivot is

$$\boldsymbol{\tau} = _____, \qquad \|\boldsymbol{\tau}\| = _____.$$

Torque measures the tendency of the force to produce rotation. Its direction gives the axis of rotation by the right-hand rule.

Commentary. Only the component of the force perpendicular to the lever arm contributes to torque.

Torque is largest when the force is _____ to the lever arm and zero when the force is _____ to it.

Example. A force $\mathbf{F} = \langle 0, 30, -40 \rangle$ N is applied at the end of a wrench whose position vector from the center of the bolt is $\mathbf{r} = \langle 0.25, 0, 0 \rangle$ m.

Find the torque vector and its magnitude. Interpret the direction.

Work:

Note. Torque and work have the same dimensional units, but they describe different quantities: torque is a vector associated with rotation, while work is a scalar.